Is It the Model or the Reading?
A Fast-Weight View of ViT Attention Maps

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Abstract

Deep-layer ViT attention maps can emphasise background patches. We test how much changing the reading rule improves these maps with the model fixed. The established fast-weight view motivates three assumptions—image specificity, spatial location and output contribution—each tested with a diagnostic proxy, correction and matched control. On DeiT-S/B, existing norm weighting raises object enrichment by $0.038/0.065$ and human alignment by $0.033/0.058$; three of four gains exceed the matched control. This reading yields no improvement exceeding our practical margin for deletion faithfulness or in DINO ViT-S. Probe relocation lowers all nine deep-layer model–metric scores relative to received attention, and combining corrections improves only one of four pre-registered comparisons. These results identify an effective alignment adjustment and an unsuccessful use of index decoding for relocation; general diagnostic prediction and the cause of relocation failure remain open.

1 Introduction

Attention maps are widely used to visualise what a vision transformer (ViT) [13] attends to [1, 8, 10]. Deep-layer maps can allocate substantial attention to high-norm background tokens; registers address this by changing the model [11, 17]. We ask how much map quality can improve by changing only the *reading rule*.

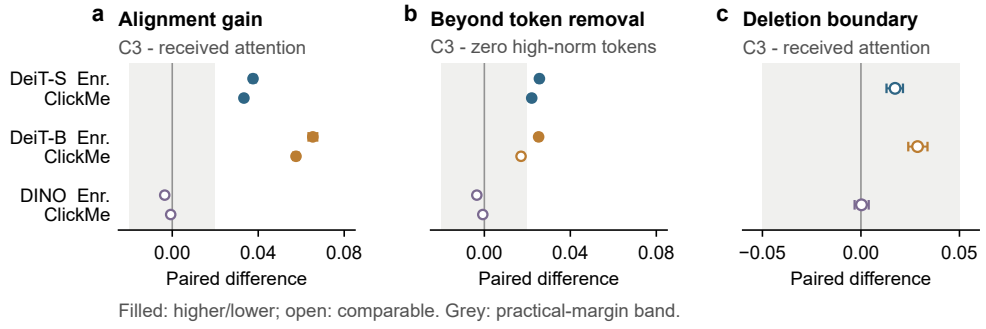


Figure 1: Contribution weighting (C3): (a) gains over received attention; (b) gains beyond zeroing high-norm tokens; (c) deletion boundary. Points/bars: original paired means/95% intervals. Filled/open: higher/lower versus comparable under the registered rule. Grey: practical-margin range (0.02; deletion 0.05).

Received attention sums attention arriving at each patch token. Interpreting it links activations to image-specific information, token indices to input locations, and activation magnitudes to output contributions. The established fast-weight view represents these tokens as hidden units. We organise tests of these links on frozen DeiT-S/B [25] and DINO ViT-S [8].

Preprint.

(1) **A controlled diagnostic framework** linking three reading assumptions to explicit proxies, corrections and controls.
(2) **Conditions and limits of existing norm weighting:** localisation and human-alignment gains in DeiT-S/B, without a deletion-margin improvement or meaningful DINO gain.
(3) **A failed proxy-guided correction:** patch-index decoding does not yield effective score relocation, and combining corrections improves only one of four registered cells.

2 Related work

2.1 Reading attention and evaluating maps. Whether attention explains predictions is debated [16, 28]. Readings incorporate layers [1], relevance propagation [9], gradients [10], norms [18, 31] or statistical filtering [3]. Kobayashi et al. already establish that activation alone omits the transformed value magnitude. We use this existing reading to test when that omission matters. Plausible maps can ignore model parameters [2], deletion depends on occlusion [5], and CNN centre priors can confound evaluation [19]; our matched controls address specific alternative explanations.

2.2 Deep tokens and fast weights. Registers absorb high-norm background tokens [11, 17], which resemble attention heads that learn to do nothing [6]; supervision also shapes ViT attention [26]. Language-model token identity can fade [7], yet deep CLS-trained ViT tokens retain similarity to their input patches [21]. Fast weights are written by another network [24]; this perspective connects linear attention to fast-weight programming [23], softmax attention to Hopfield memory [22], and motivates efficient attention such as MiTA [27]. We use it to organise an empirical study of reading assumptions, rather than propose a new attention mechanism.

3 Method

3.1 An exact representation, an interpretive framework. In layer ℓ, normalised tokens $x_1, \ldots, x_{N+1}$ (CLS and $N = 196$ patches) enter H heads. Head h forms $q_i^h = W_Q^h x_i$, $k_j^h = W_K^h x_j$ and $v_j^h = W_V^h x_j$; its attention update is

$$o_i = \sum_h W_O^h \sum_j a_{ij}^h v_j^h, \qquad a_{ij}^h = \mathrm{softmax}_j\big(q_i^{h\top} k_j^h / \sqrt{d}\big). \tag{1}$$

For a fixed image, each head is exactly a two-layer network in q_i^h: hidden unit j has input weights k_j^h, activation a_{ij}^h and output weights $W_O^h v_j^h$. The softmax couples hidden activations through normalisation. This algebraic representation is exact; the reading assumptions derived from it require empirical tests. Received attention is $s_j = \sum_{h,i} a_{ij}^h$, placed at patch j. Queries include CLS and all patches; maps exclude CLS and sum layers 9–12, fixed before the formal runs.

3.2 Assumptions, proxies and controls. Fig. 2 links three assumptions to tests: shared structure may obscure image-specific information; unit meaning requires measurement [4]; and activation omits output weights [18, 30]. These are interpretive assumptions; diagnostics remain proxies. Each correction must improve over received attention *and* exceed its matched control to support an assumption-specific benefit. Contribution weighting measures output-vector magnitudes, omitting cancellation and later computation. Probe relocation uses $P(p \mid x_j)$ trained to decode token index p, not input-content provenance.

Corrections are $s - \bar{s}$, $r_p = \sum_j s_j P(p \mid x_j)$ and $c_j = \sum_{h,i} a_{ij}^h \|W_O^h v_j^h\|$. Their controls subtract a centred Gaussian, blur by matched displacement and zero high-norm tokens, respectively.

3.3 Metrics and decision rules. A centred Gaussian correlates 0.41 with raw ClickMe maps. Two *centre-controlled* metrics address this prior: *same-ring enrichment*, the object rate of top-scoring patches minus the rate at the same radius (ImageNet-S masks [15]); and *image-specific ClickMe alignment*, Spearman correlation with the image's human map minus that with another image's [14, 20]. These controls target specific prior effects. *Deletion faithfulness* compares least- and most-important-first token-removal curves [5]. With 1,000 paired image-bootstrap resamples, a difference is *higher/lower* if its 95% interval excludes zero and its magnitude exceeds 0.02 (0.05 for deletion); otherwise it is *comparable*, a registered category rather than formal equivalence.

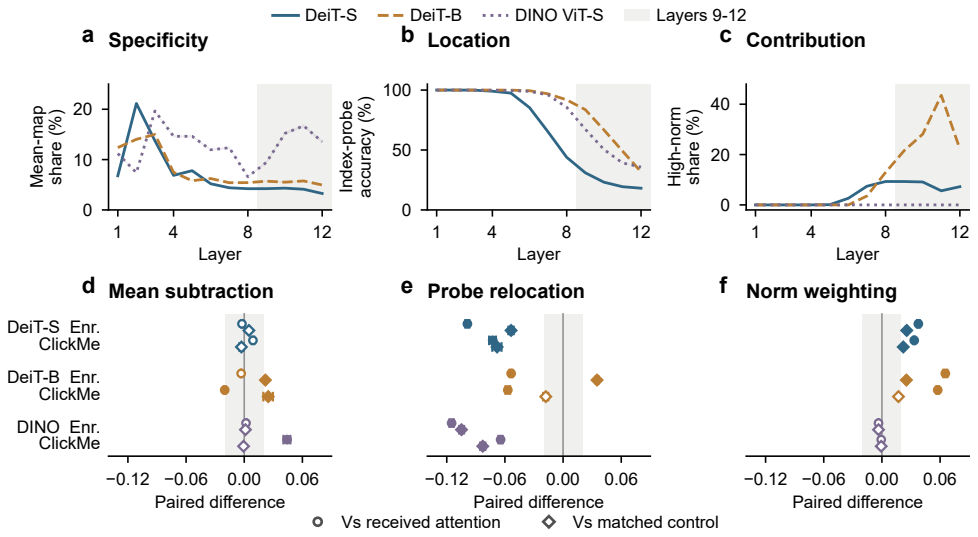


Figure 2: Diagnostic proxies (a–c) and corresponding corrections (d–f), matched by column. Points/bands/bars: original means and 95% intervals. Circle/diamond: versus received attention/matched control. Filled/open: higher/lower versus comparable. Grey: layers 9–12 (top); ± 0.02 margin (bottom).

4 Experiments

4.1 Setup and registered predictions. We evaluate frozen supervised DeiT-S/B and DINO ViT-S/16. Model-side comparisons use a test-time register [17] and the released MiTA top-k checkpoint [27], both at size S. Data comprise 5,000 ImageNet images [12], 12,419 ImageNet-S images, 17,408 ClickMe maps and 4,000 held-out images (2,000 probe-train; 2,000 calibration/test). Metrics, margins and comparisons were fixed before formal runs (App. A). **R1:** contribution weighting in DINO and relocation at layers 1–4 remain comparable. **R2:** all three corrections improve all four supervised centre-controlled cells. **R3:** neither model-side variant is higher than DeiT-S. Single-correction/control comparisons had no registered direction; their diagnosis–benefit interpretation is post hoc.

4.2 Execution and sanity checks. Restricting attention to $k = 16$ region-selected units preserves $72/83/61\%$ of predictions in DeiT-S/B/DINO, versus $9/30/4\%$ for random units (App. B). Existing parameter-randomisation checks pass for all five readings under their registered criterion (App. F).

4.3 Diagnostic patterns (Fig. 2). The mean map explains only 3–6% of spatial variance in supervised layers 6–12, at most 17% in DINO; in layers 9–12 its structure is 82–98% radial. Index-probe accuracy declines from nearly 100% in early layers to $18/32/36\%$ at layer 12 (chance: 0.5%). High-norm tokens (block-input norm > 3 times the image median) receive up to $9/44\%$ of attention in DeiT-S/B and none in DINO. Their value norms average $0.29/0.23$ times the other patches' median in deep layers (App. B). These proxies motivate tests; they do not themselves establish misreading.

4.4 Corrections and registered outcomes (Table 1). Contribution weighting improves alignment, with limits. Enrichment rises by $0.038/0.065$ and ClickMe by $0.033/0.058$ in DeiT-S/B. Gains exceed the zero-high-norm control in three of these four cells; DeiT-B ClickMe does not. Deletion gains ($0.017/0.029$) are below the 0.05 margin. DINO differences are at most 0.004 in magnitude, satisfying the contribution part of R1. Post hoc, correctly assigned weights outperform shuffled weights in all four supervised cells; value-only weighting is comparable (App. B). Thus the useful correction is an existing norm-based reading [18, 31].

Other corrections do not yield consistent benefits. Mean subtraction's only R0 gain (DINO ClickMe) matches its control. Relocation lowers all nine deep-layer cells and underperforms matched blur in DeiT-S and DINO. Early relocation remains comparable (R1). The combination improves only DeiT-B ClickMe: R2 *partly holds* (1/4), without consistent benefit.

Table 1: Layers 9–12; all metrics ↑. Arrows: higher/lower versus received attention under the registered rule (model-side rows: DeiT-S). ◇: higher than the matched control. **Bold**: higher than both. Controls are indented. GAE [10] uses gradients and is a descriptive reference. Enr.: enrichment; Click.: ClickMe; Del.: deletion.

	DeiT-S			DeiT-B			DINO ViT-S		
Reading	Enr.	Click.	Del.	Enr.	Click.	Del.	Enr.	Click.	Del.
Received attention	0.140	0.157	0.236	0.080	0.107	0.169	0.199	0.208	0.285
Contribution weighting	**0.178**$^{\uparrow\diamond}$	**0.190**$^{\uparrow\diamond}$	0.253	**0.146**$^{\uparrow\diamond}$	0.165$^{\uparrow}$	0.198	0.196	0.207	0.285
zero high-norm tokens	0.152	0.168	0.241	0.121$^{\uparrow}$	0.148$^{\uparrow}$	0.186	0.199	0.208	0.285
Mean-map subtraction	0.138	0.165	0.208	0.077$^{\diamond}$	0.087$^{\downarrow\diamond}$	0.144$^{\diamond}$	0.201	0.252$^{\uparrow}$	0.233$^{\downarrow}$
subtract centre Gaussian	0.133	0.168	0.167$^{\downarrow}$	0.056$^{\downarrow}$	0.062$^{\downarrow}$	0.024$^{\downarrow}$	0.200	0.253$^{\uparrow}$	0.232$^{\downarrow}$
Probe relocation	0.041$^{\downarrow}$	0.084$^{\downarrow}$	0.076$^{\downarrow}$	0.027$^{\downarrow\diamond}$	0.050$^{\downarrow}$	0.051$^{\downarrow}$	0.084$^{\downarrow}$	0.143$^{\downarrow}$	0.141$^{\downarrow}$
matched blur	0.095$^{\downarrow}$	0.152	0.181$^{\downarrow}$	−0.008$^{\downarrow}$	0.068$^{\downarrow}$	0.097$^{\downarrow}$	0.189	0.226	0.287
All three corrections	0.059$^{\downarrow}$	0.123$^{\downarrow}$	0.076$^{\downarrow}$	0.079	0.137$^{\uparrow}$	0.085$^{\downarrow}$	0.088$^{\downarrow}$	0.177$^{\downarrow}$	0.090$^{\downarrow}$
+ test-time register	0.152	0.169	0.236	–	–	–	–	–	–
Top-k training (MiTA)	0.100$^{\downarrow}$	0.115$^{\downarrow}$	–	–	–	–	–	–	–
GAE (reference)	0.204	0.234	0.345	0.196	0.213	0.308	0.221	0.234	0.418

Model-side results have a narrower scope. The register gains $0.012/0.013$ on enrichment and ClickMe, below the 0.02 margin. MiTA scores lower, but differs in training and preprocessing. R3 holds under the registered comparisons; these results do not establish a general advantage over model-side interventions.

4.5 Does diagnosis predict correction benefit? The post-hoc correspondence holds for contribution weighting, not relocation. Token-specific weights matter, but dose–gain associations are weak (Spearman 0.045–0.158; App. B). Local errors and the tested MLP probe narrow relocation explanations (App. C); probe/calibration limitations remain possible. Index decodability differs from content provenance: local content may persist after position codes weaken [21]. Whether that mismatch causes relocation failure remains untested.

5 Discussion

5.1 Interpretation and limitations. Reading changes can improve localisation and human alignment without improving deletion beyond the chosen margin. A declining diagnostic is therefore insufficient grounds for a correction. Evidence covers three frozen S/B classifiers and received attention only; CLS attention, larger/register-trained models and repeated training seeds are untested. The high-norm threshold was not varied. ClickMe images likely overlap model training data. Gains are modest, visually subtle (App. D) and below GAE's; diagnostic prediction beyond these checkpoints remains open.

5.2 Ethical considerations. Public images and checkpoints contain people and known biases. Improved map alignment remains partial evidence about model behaviour.

6 Conclusion

Existing norm weighting improves deep-layer localisation/alignment in DeiT-S/B; probe relocation fails and the registered combination lacks consistent gains. The fast-weight view organises these controlled tests. Validating diagnostic-guided correction requires token-to-input attribution and broader model coverage. Reading changes should be compared with matched controls, while separating alignment from prediction faithfulness.